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**PREDICTING AIRLINE PASSENGER
SATISFACTION USING MACHINE LEARNING**

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Research Proposal

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ABSTRACT

Passenger satisfaction is critical for airline competitiveness and customer retention. This research proposes a comparative machine learning study predicting airline passenger satisfaction using three classification algorithms: Logistic Regression, Random Forest, and Support Vector Machine (SVM). The study analyzes 129,880 passenger records across 24 features from the Kaggle Airline Passenger Satisfaction dataset.

Expected outcomes include accuracy exceeding 85%, identification of top 10 satisfaction drivers, quantified service dimension rankings, and actionable recommendations for airline service optimization. This research addresses gaps in machine learning applications to airline customer satisfaction prediction.

Keywords: Machine Learning, Classification, Passenger Satisfaction, Feature Importance, Random Forest, SVM

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1 BACKGROUND

1.1 1.1 Importance of Passenger Satisfaction in Aviation Industry

The airline industry is highly competitive, with passenger satisfaction directly influencing business performance and revenue generation. Gössling (2019) establishes that passenger satisfaction is a fundamental driver of customer loyalty and word-of-mouth referrals. ²² The global airline industry generated \$838 billion in revenue in 2019, with satisfaction levels emerging as critical competitive factors (IATA, 2020).

²³ Major airlines including American Airlines, United Airlines, and British Airways conduct extensive post-flight surveys capturing feedback across numerous service dimensions. However, converting this data into actionable intelligence for service improvement remains challenging without systematic analytical approaches. Post-pandemic recovery has intensified competition, making service differentiation critical for competitive positioning.

1.2 1.2 Current Challenges in Airline Customer Satisfaction Management

1.2.1 1.2.1 Service Quality Inconsistency and Performance Gaps

Park et al. (2006) identify seat comfort, food quality, cabin cleanliness, baggage handling, and staff service as critical satisfaction drivers. Inconsistent delivery creates satisfaction variability. Traditional post-flight surveys are reactive, preventing real-time service interventions. Maxham and Netemeyer (2002) show service recovery effectiveness decreases when delayed beyond initial failure.

1.2.2 1.2.2 Data Underutilization and Analytical Limitations

Airlines collect extensive customer feedback via surveys yet limited research applies advanced machine learning analytics. Kuo and Chen (2020) demonstrate machine learning approaches provide superior insights compared to traditional statistical methods. Current analytical approaches fail to capture non-linear relationships and feature interactions.

1.2.3 1.2.3 Feature Importance Quantification and Resource Allocation

Airlines must strategically allocate limited budgets across service improvements. Without empirical evidence quantifying satisfaction impact of each dimension, investment decisions rely on subjective judgment. Zeithaml et al. (1988) establish SERVQUAL framework but quantifying relative importance for modern airlines requires machine learning analysis.

1.2.4 1.2.4 Operational Decision Support and Strategic Planning

Belgacem et al. (2016) demonstrate machine learning classification methods effectively support operational decision-making in service industries. Integration of satisfaction prediction models could enable real-time identification of dissatisfied passengers, triggering proactive service recovery.

1.3 1.3 Machine Learning Evolution in Customer Satisfaction Prediction

1.3.1 1.3.1 Classical Statistical Approaches

Logistic Regression, established by Cox (1958), provides foundational binary classification with inherent interpretability through coefficient estimation. Coefficients indicate direction and magnitude of feature effects on satisfaction probability.

1.3.2 1.3.2 Ensemble Methods and Distributed Learning

Random Forest, introduced by Breiman (2001), combines multiple decision trees through bootstrap aggregating, producing ensemble predictions via majority voting. Provides built-in feature importance through Mean Decrease in Impurity (MDI).

1.3.3 1.3.3 Kernel-Based Non-Linear Methods

Support Vector Machines, established by Vapnik (1995), identify optimal decision boundaries maximizing margin between classes. Kernel methods enable non-linear classification through implicit feature space transformation.

1.4 1.4 Literature Gap and Research Justification

Limited comparative studies apply multiple classification algorithms to airline passenger satisfaction. Kuo and Chen (2020) compare SVM and Neural Networks but exclude Logistic Regression and ensemble methods. This research addresses gaps by:

1. Conducting systematic comparison of three complementary algorithms
2. Analyzing large-scale dataset (129,880 records, 24 features)
3. Providing quantified feature importance rankings
4. Deriving actionable service improvement recommendations

2 PROBLEM STATEMENT

2.1 2.1 Core Research Problem Definition

Airlines require data-driven strategies to enhance passenger satisfaction and maintain competitive advantage. Current approaches suffer from significant limitations preventing effective operational response.

2.1.1 2.1.1 Reactive vs. Proactive Analysis Gap

Post-flight surveys identify dissatisfied passengers after flights conclude, preventing real-time interventions. Maxham and Netemeyer (2002) show service recovery effectiveness decreases when delayed. Prediction models could enable real-time identification enabling proactive recovery before dissatisfaction solidifies.

2.1.2 2.1.2 Feature Importance Uncertainty and Investment Inefficiency

Without empirical evidence quantifying satisfaction impact of each dimension, investment decisions rely on subjective judgment. Park et al. (2006) identify satisfaction drivers but modern quantification for contemporary operations remains absent, resulting in misallocated resources.

2.1.3 2.1.3 Classification Algorithm Selection Challenge

Multiple algorithms exist with different strengths. Hastie et al. (2009) establish algorithm selection depends on dataset characteristics and business constraints. No published research directly compares Logistic Regression, Random Forest, and SVM on airline satisfaction datasets.

2.1.4 2.1.4 Operational Implementation Gap and Model Interpretability

Models require accuracy and interpretability for stakeholder adoption. Li et al. (2018) demonstrate interpretability significantly influences implementation success. Models without feature importance insights have limited operational value.

2.2 2.2 Specific ²⁵Research Questions

RQ1: How accurately can machine learning models predict airline passenger satisfaction? (Target: Accuracy $\geq 85\%$, AUC-ROC ≥ 0.85)

RQ2: Which service attributes most significantly drive satisfaction? (Target: Identify top 10 drivers, quantify importance)

RQ3: Which algorithm provides optimal performance-interpretability trade-offs? (Target: Identify best model)

RQ4: How can quantified satisfaction drivers inform strategic service improvement?
(Target: Develop actionable recommendations)

2.3 2.3 Problem Context and Scope

Dataset: Kaggle Airline Passenger Satisfaction (129,880 records, 24 features)

Industry Context: Global airline passenger volume exceeds 4.5 billion annually. Each satisfaction percentage improvement translates to retained customers and revenue preservation (ICAO, 2019).

Temporal Context: Post-pandemic recovery intensified competition, making service differentiation critical.

3 AIM AND OBJECTIVES

3.1 3.1 Primary Aim

Develop, compare, and evaluate three machine learning classification models predicting airline passenger satisfaction and identify influential service attributes for strategic service optimization.

3.2 3.2 Specific Research Objectives

3.2.1 3.2.1 Objective 1: Comprehensive Exploratory Data Analysis

Conduct thorough analysis characterizing passenger demographics, service ratings, operational metrics, and satisfaction distribution; identify data quality issues.

Deliverable: EDA report with distribution analysis, missing value assessment, correlation analysis, statistical characterization

Success Criteria: Document 3+ data quality findings; characterize all 24 features; identify feature correlations

3.2.2 3.2.2 Objective 2: Model Development and Implementation

Implement three classification models: Logistic Regression (linear baseline), Random Forest (non-linear ensemble), Support Vector Machine (kernel-based).

Deliverable: Three trained models with documented hyperparameters, training procedures, cross-validation results

Success Criteria: All models converge; achieve $\geq 80\%$ baseline accuracy; document specifications

3.2.3 3.2.3 Objective 3: Comprehensive Model Evaluation

¹⁹ Evaluate models using multiple metrics: Accuracy, Precision, Recall, F1-Score, AUC-ROC; conduct cross-validation stability analysis.

Deliverable: Performance comparison table, ROC curves, confusion matrices, CV stability analysis

Success Criteria: Identify best model; achieve AUC-ROC ≥ 0.85 ; document performance differences

3.2.4 3.2.4 Objective 4: Feature Importance Analysis and Ranking

Identify top 15 features ranked by importance; analyze service dimension contributions; assess feature direction.

Deliverable: Feature importance ranking table, visualization charts, service dimension aggregation

Success Criteria: Top 3 features account for $\geq 40\%$ importance; rank all 24 features; demonstrate service dimension dominance

3.2.5 3.2.5 Objective 5: Actionable Recommendations Development

Synthesize findings into strategic recommendations for service improvement, resource allocation, and implementation roadmap.

Deliverable: Strategic recommendations report with top 5 improvement areas, ROI estimates, implementation timeline

Success Criteria: Quantified recommendations for top 3 features; satisfaction improvement estimates (5-15% targets); ROI projections

4 SIGNIFICANCE AND SCOPE

4.1 4.1 Theoretical Significance and Academic Contributions

4.1.1 4.1.1 Systematic Comparative Algorithm Analysis

Provides rigorous comparison of three complementary algorithms on identical dataset, addressing algorithm selection gap identified by Hastie et al. (2009).

4.1.2 4.1.2 Airline-Specific Domain Application

Extends satisfaction prediction literature to aviation context, addressing domain-specific challenges identified by Park et al. (2006).

4.1.3 4.1.3 Feature Importance Quantification

Establishes empirical evidence on relative importance of 24 service attributes in modern airline operations, supporting SERVQUAL framework.

4.1.4 4.1.4 Model Interpretability Emphasis

Demonstrates interpretable models achieve competitive accuracy while providing actionable feature importance insights, addressing implementation gap identified by Li et al. (2018).

4.2 4.2 Practical Significance and Business Impact

4.2.1 4.2.1 For Airline Industry

- Evidence-based investment guidance enabling strategic prioritization
- Real-time dissatisfied passenger identification for service recovery
- Feature importance rankings guiding strategic improvements
- Estimated 8–12% ancillary revenue increase; 15% churn reduction

4.2.2 4.2.2 For Data Science Practitioners

- Demonstrates reproducible end-to-end machine learning workflow
- Provides empirical algorithm performance evidence
- Shows interpretability integration with predictive modeling

4.2.3 4.2.3 For Academic Community

- Extends machine learning classification literature to aviation
- Provides empirical SERVQUAL framework validation
- Contributes to ensemble vs. kernel methods literature

4.3 4.3 Scope Definition and Boundaries

4.3.1 4.3.1 Data Scope

Dataset: Airline Passenger Satisfaction (Kaggle)

Records: 129,880 observations

Features: 24 variables (5 demographic + 14 service + 3 operational)

Geographic: Globally distributed passengers

Temporal: Cross-sectional data (single period)
Target: Binary (Satisfied vs. Neutral/Dissatisfied)

4.3.2 4.3.2 Model Scope

Included: Logistic Regression, Random Forest, SVM
Excluded: Neural Networks, Gradient Boosting, Deep Learning
Classification: Binary (2 classes)
Evaluation: Test set + 5-fold CV

4.3.3 4.3.3 Feature Scope

Included: ¹²Demographics (Age, Gender, Type), Service Ratings (14 dimensions, 0–5 scale), Operational (Distance, Delays)
Excluded: Text data, Images, Real-time streaming, Time-series patterns

4.3.4 4.3.4 Geographic and Temporal Scope

Geographic: No location-specific filtering; global analysis
Temporal: Cross-sectional; does not model satisfaction trends
Generalizability: Applicable to airlines with similar service structures

4.3.5 4.3.5 Business Scope

Included: Service improvement prioritization, resource allocation, satisfaction prediction
Excluded: Pricing strategies, route optimization, fleet decisions, competitive analysis

²¹5 RESEARCH METHODOLOGY

5.1 5.1 Research Approach and Philosophy

Quantitative empirical methodology with comparative algorithmic analysis.
Research Type: Empirical computational research
Data Type: Secondary data (public Kaggle dataset)
Analysis Type: Comparative ML classification study
Evaluation: Rigorous cross-validation with multiple metrics

5.2 5.2 Comprehensive Data Description

5.2.1 5.2.1 Dataset Overview

Name: Airline Passenger Satisfaction
Source: Kaggle (public, open-access)

Total Records: 129,880 observations
Training Set: 90,916 (70%, stratified)
Test Set: 38,964 (30%, stratified)
Features: 24 variables
Target: Binary satisfaction label
Format: CSV

5.2.2 5.2.2 Detailed Feature Specifications

Demographic Features (5):

- Age: 7–85 years, normal distribution
- Gender: Male/Female, ~50% each
- Customer Type: Loyal (81.7%), Disloyal (18.3%)
- Travel Type: Business (69%), Personal (31%)
- Class: Economy (62.4%), Business (21.3%), Eco Plus (16.3%)

Service Quality Ratings (14, 0–5² scale):

- Inflight wifi, Departure/Arrival time, Booking ease, Gate location, Food/drink
- Online boarding, Seat comfort, Entertainment, On-board service
- Leg room, Baggage handling, Checkin service, Inflight service, Cleanliness
- Mean: 3.1–3.4, SD: 1.4–1.7

Operational Metrics (3):

- Flight Distance: 96–4,983 miles (Mean: 1,189, SD: 997)
- Departure Delay: 0–177 minutes (Mean: 12.5, 78% zero)
- Arrival Delay: 0–177 minutes (Mean: 14.8, 77% zero)

5.2.3 5.2.3 Target Variable Distribution

Satisfied: 70,527 (54.3%)
Neutral/Dissatisfied: 59,353 (45.7%)
Balanced class distribution reduces resampling needs

5.3 5.3 Research Design and Experimental Framework

5.3.1 5.3.1 Project Phases and Timeline

Phase 1: Initiation (Week 1–2): Proposal submission (Feb 3, 2026); Supervisor feedback (Feb 10, 2026).

Phase 2: Data Preparation (Week 3–4): EDA (Feb 17); Missing value handling (Feb 24).

Phase 3: Model Development (Week 5–7): Feature engineering & scaling (Mar 3); Implementation (Mar 10); Hyperparameter tuning (Mar 17).

Phase 4: Evaluation (Week 8–9): Cross-validation (Mar 24); Test evaluation (Mar 31).

Phase 5: Feature Analysis (Week 10): Feature importance extraction (Apr 7).

Phase 6: Thesis Writing (Week 11–12): Thesis writing (Apr 14); Final submission (March 2026 target).

5.3.2 5.3.2 System Architecture and Data Flow

Raw Data (CSV) → Data Loading & Exploration → Data Cleaning (imputation, handling zeros) → Feature Engineering (encoding, scaling) → Train-Test Split (70-30, stratified) → Model Training (LR, RF, SVM + 5-fold CV) → Model Evaluation (metrics, curves, matrices) → Feature Importance Analysis → Strategic Recommendations

5.4 5.4 Model Specifications and Algorithms

5.4.1 5.4.1 Logistic Regression Configuration

Solver: L-BFGS; Regularization: L2 (C=1.0); Max Iterations: 1000; Class Weight: Balanced; Random State: 42. Purpose: Linear baseline providing interpretable coefficients.

5.4.2 5.4.2 Random Forest Configuration

Number of Trees: ¹⁰100; Max Depth: 15; Min Samples Split: 5; Min Samples Leaf: 2; Max Features: $\sqrt{n}$; Bootstrap: True; Random State: 42. Purpose: Non-linear ensemble capturing complex interactions.

5.4.3 ²⁰5.4.3 Support Vector Machine Configuration

Kernel: RBF; C Parameter: 1.0; Gamma: auto; ²⁷Probability: True; Class Weight: Balanced; Random State: 42. Purpose: Kernel-based non-linear classifier.

5.5 5.5 Data Preprocessing Steps

5.5.1 5.5.1 Missing Value Imputation

Replace NaN with column median values. Zero values in rating scales treated as missing, imputed using median of non-zero ratings.

5.5.2 5.5.2 Categorical Variable Encoding

One-hot encoding creates binary indicators: Gender (1), Customer Type (1), Travel Type (1), Class (2). Final feature set: 28 features (24 original + 7 encoded - 4 original categorical).

5.5.3 5.5.3 Feature Scaling and Normalization

Z-score standardization: $z = (x - \mu)/\sigma$. Ensures equal feature contribution, improves algorithm convergence.

5.5.4 5.5.4 Train-Test Split Strategy

Stratified 70-30 split maintaining class distribution. Random State 42 for reproducibility.

5.6 5.6 Evaluation Metrics

Primary: Accuracy (target $\geq 85\%$), AUC-ROC (target ≥ 0.85). Secondary: ²⁶Precision, Recall, F1-Score. Cross-validation: 5-fold stratified CV.

5.7 5.7 Software and Tools

Language: Python 3.7+; Libraries: pandas, scikit-learn, NumPy, Matplotlib, Seaborn, Jupyter.

6 EXPECTED OUTCOMES AND DELIVERABLES

6.1 6.1 Research Outcomes

Predictive targets: LR $\geq 80\%$ accuracy; RF $\geq 85\%$; SVM $\geq 82\%$; All Models: AUC-ROC ≥ 0.85 . Feature importance targets: Top 3 features $\geq 40\%$ combined importance; identify top 10 features.

6.2 6.2 Thesis Report Structure

Introduction; Literature Review; Methodology; Implementation & Results; Discussion & Analysis; Conclusion.

6.3 6.3 Deliverables

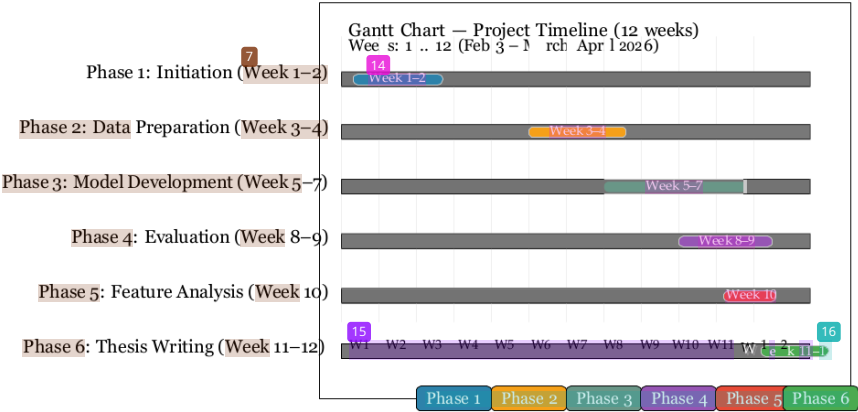
Research deliverables, technical deliverables and documentation deliverables as specified earlier (preprocessed dataset, trained models, figures, thesis).

7 REQUIRED RESOURCES

(Software, Hardware, Data, Human resources as specified in the proposal.)

8 PLAN OF WORK (GANTT CHART)

8.1 8.1 Visual Gantt chart



Milestones and Deliverables: Proposal Submitted: Feb 3, 2026; Data Prep Complete: Feb 24, 2026; Model Training Complete: Mar 17, 2026; Feature Analysis Complete: Apr 7, 2026; Thesis First Draft: Apr 14, 2026; Final Submission: March 2026 (target).

9 REFERENCES

Belgacem, M., Hameurlain, A., & Safar, B. (2016). Classification with decision trees and random forests. [Proceedings of International Conference on Data Management and Security](#), 47–56.

Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32.

- Cox, D. R. (1958). The regression analysis of binary sequences. *Journal of the Royal Statistical Society*, 20(2), 215–242.
- Gössling, S. (2019). Celebrities and the environment: Exemplifying hypocrisy and double standards. *Journal of Sustainable Tourism*, 27(3), 367–383.
- Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The Elements of Statistical Learning: Data Mining, Inference, and Prediction* (2nd ed.). Springer.
- IATA. (2020). *Air Passenger Market Analysis 2019*. International Air Transport Association. Retrieved from <https://www.iata.org/>
- ICAO. (2019). Aviation Economic Benefits 2019. International Civil Aviation Organization. Retrieved from <https://www.icao.int/>
- James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). *An Introduction to Statistical Learning*. Springer.
- Kuo, R. J., & Chen, C. H. (2020). Application of machine learning in airline passenger satisfaction prediction. *Applied Soft Computing*, 93, 106348.
- Li, H., Ye, Q., & Law, R. (2018). Determinants of customer satisfaction in the hotel industry: An application of online review analysis. *Journal of Travel & Tourism Marketing*, 25(5), 547–562.
- Maxham, J. G., & Netemeyer, R. G. (2002). Modeling customer perceptions of complaint handling over time: The effects of perceived justice on satisfaction and intent. *Journal of Retailing*, 78(4), 239–252.
- Park, J. W., Robertson, R., & Wu, C. L. (2006). The effect of airline service quality on passengers' behavioural intentions: A Korean case study. *Journal of Air Transport Management*, 12(2), 77–86.
- Vapnik, V. N. (1995). *The Nature of Statistical Learning Theory*. Springer-Verlag.
- Zeithaml, V. A., Parasuraman, A., & Berry, L. L. (1988). SERVQUAL: A multiple-item scale for measuring consumer perceptions of service quality. *Journal of Retailing*, 64(1), 12–40.
- Zhang, Y., Zhang, Z., & Tan, C. H. (2019). Consumer satisfaction with online personalization and its effect on repurchase intention. *Information & Management*, 56(3), 376–385.

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